

Data Vortex A'26 - Round 3 Analytical Report

Public Reaction to the CM Vijay – Actress Trisha Controversy

1. Data Collection Method

- **Source & pipeline:** Data was collected by *scraper.py* (Round 3 deliverable) into *live_dataset.csv*, with a single source in this pull: **YouTube comments** (*--sources youtube --yt-order relevance*), which surfaces the most-commented videos matching the query terms rather than strictly the newest ones.
- **Volume before cleaning:** 8,854 raw records spanning 2017-08-28 to 2026-09-22.
- **Cleaning steps applied in the notebook:**
 1. Parsed `created_utc` to UTC timestamps; dropped rows with missing timestamp/text.
 2. De-duplicated on (source, post_id).
 3. **Window restriction (critical step):** Because "most-commented" pulled in unrelated legacy content (old *Ghilli* song uploads, years-old Vijay/Trisha fan edits from 2017–2018), the notebook filters to `created_utc >= 2026-08-20`. This dropped 6,860 off-topic/legacy records, leaving **1,994 records** genuinely tied to the live controversy window.
 4. **Language flagging:** a regex-based classifier tags each post as en, ta (Tamil script), mixed, or tanglish (Latin-script Tamil), since the Round 2 sentiment/topic models and VADER are English-only.
 - en: 68.5% (1,366) | ta: 23.6% (471) | mixed: 4.0% (80) | tanglish: 3.9% (77)
 - 628 records (31.5%) required translation before modeling.
 5. **Translation:** non-English/Tanglish text passed through googletrans to produce `translated_text`, manually spot-checked against a 100-row translation-quality audit file and a 15-sample manual review.
 6. **Text normalization:** `repair_text` → `clean_text` → `meta_features` pipeline from the Round 2 module prepares text for the pickled TF-IDF + LinearSVC (calibrated) sentiment and topic models.
- **Limitation to flag:** Only one platform (YouTube) is represented no Twitter/X, Instagram, or news-comment data so the sample skews toward video-comment discourse and cannot be read as fully representative of overall public sentiment. Also, Tanglish text passed the en gate but was never part of the model's training distribution, so en-labeled sentiment/topic predictions on Tanglish rows should be treated as lower-confidence.

2. Time Window

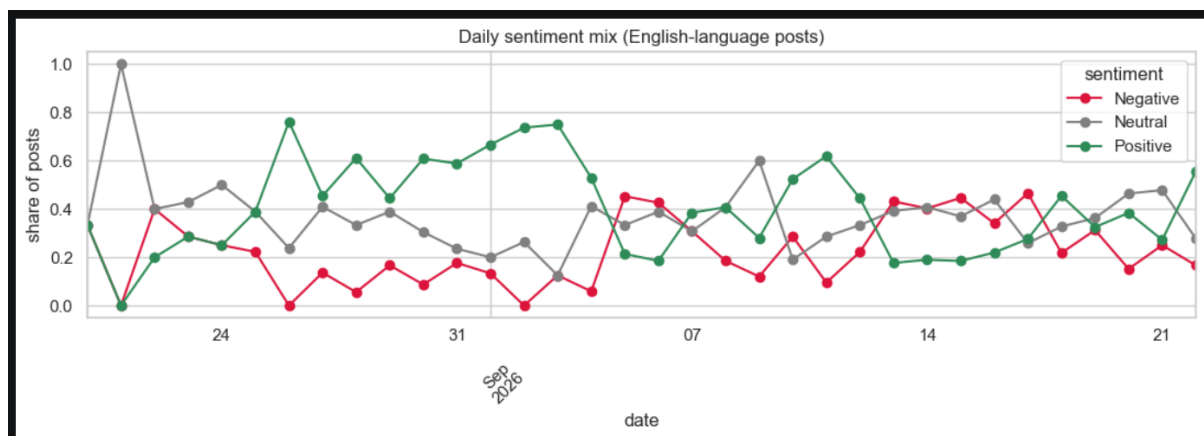
- **Full collection window analyzed:** 2026-08-20 07:18 UTC → 2026-09-22 12:41 UTC (≈33 days), after the legacy-content filter.
- **Granularity:** Activity was bucketed both **hourly** (fine-grained, for the live 40-hour collection push) and **daily** (for the full window), matching the two spike-detection passes in the notebook.
- **Distribution shape:** Volume is low and flat from Aug 20–Sep 4 (roughly 1–23 posts/day), then punctuated by three sharp daily bursts Sep 5, Sep 13–14, and Sep 21 before tapering by Sep 22.

3. Sentiment Analysis

- **Method:** A pickled TF-IDF + calibrated LinearSVC classifier (from the Round 2 pipeline) labels each English-language post Positive / Negative / Neutral. Only lang == "en" rows (1,366 of 1,994) received a label; Tamil/mixed rows were excluded from the model but retained for volume/entity counts.
- **Overall distribution (English subset):** Neutral 776 (39%), Negative 645 (32%), Positive 573 (29%) a genuinely mixed, contested conversation rather than a one-sided pile-on.
- **Shift detection method:** Day-over-day change in the *negative-sentiment share* is compared against a threshold of 0.15 (15 percentage points). **17 days** crossed this threshold across the window well above the "at least 2" requirement.

Two representative sentiment shifts (with the required minimum of ≥ 2):

Date	Negative share	Δ Negative	Δ Positive	Reading
2026-09-05	45.3% (up from ~5.9% prior day)	+39.4 pts	-31.6 pts	Sharpest negative swing in the dataset coincides with the first major volume spike (117 posts that day, entity mentions of "vijay" jumping to 65 and "trisha" to 49, plus first appearance of "dmk"/"sangeetha" terms at volume).
2026-09-13	43.1% (up from ~9.5% two days prior)	+21.0 pts	-26.8 pts	Second negativity spike, right at the start of the largest activity burst of the whole window (255 posts that day, rising to 395 the next day).
(secondary) 2026-08-22	40.0%	+40.0 pts	+20.0 pts	Early, low-volume (5 posts) swing flagged statistically but should be treated cautiously given the tiny sample size that day.



Sentiment analysis and categorisation on a daily basis

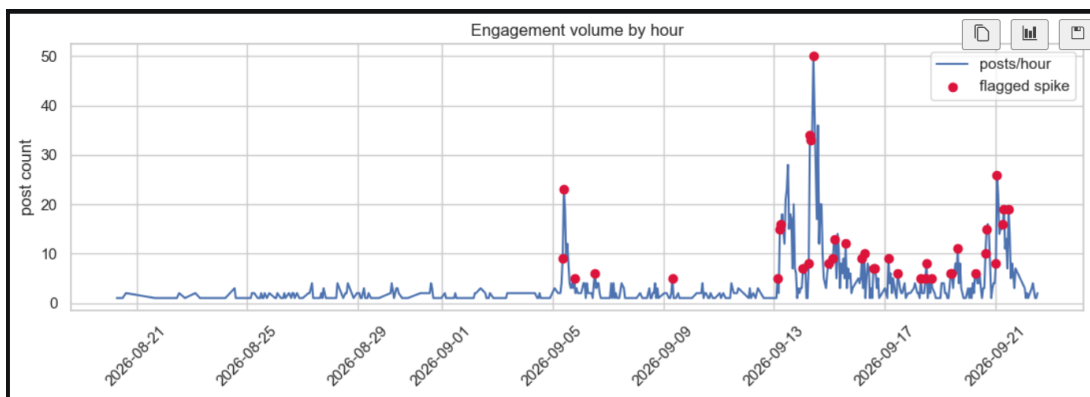
- **Interpretation:** The two large, high-volume shifts (Sep 5 and Sep 13) both move sentiment *negative* and both land right at the front edge of a volume spike i.e., new information entering the conversation pushed both attention **and** tone negative simultaneously, rather than negativity building up gradually. Positive share never recovers to its pre-controversy level (mid-40s%→low-20s%) after Sep 5, suggesting the story hardened public opinion rather than blowing over.

4. Activity Analysis

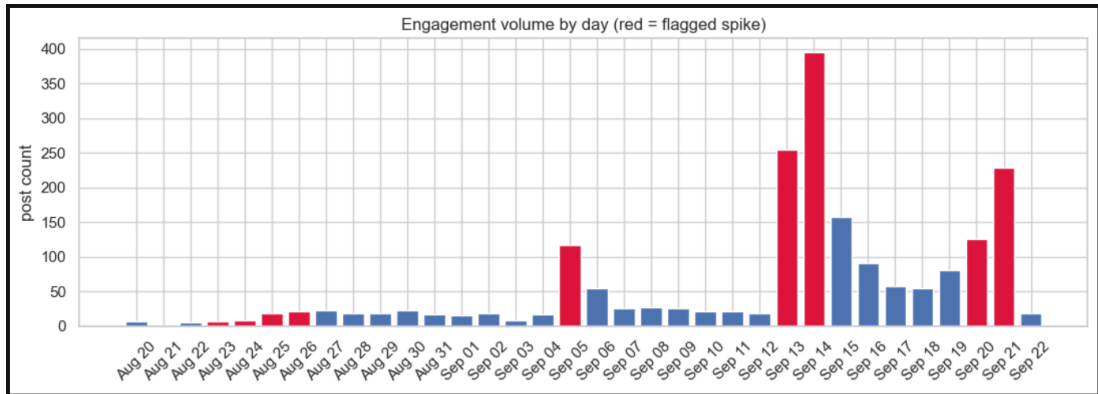
- **Method:** Post counts bucketed hourly and daily; a bucket is flagged as a **spike** if volume ≥ 5 posts *and* is either $\geq 1.75\times$ the rolling 3-bucket baseline mean, or ≥ 2 standard deviations above it (previous-3-bucket window, excluding the current bucket so a spike doesn't dilute its own signal).
- **Results:** 38 hourly spikes and multiple daily spikes were flagged comfortably exceeding the "at least 1 engagement spike" requirement.

Headline engagement spike:

- **2026-09-14, 07:00–10:00 UTC:** hourly volume jumped from a baseline of $\sim 3\text{--}14$ posts/hour to **34, then 33, then 50 posts/hour** the single largest sustained burst in the dataset, and part of the record daily total of **395 posts on Sep 14** (compared to a pre-controversy daily baseline in the teens/twenties).
- Two further large hourly spikes: **2026-09-05, 10:00 UTC** (23 posts vs. baseline 5) and **2026-09-21, 01:00 UTC** (26 posts vs. baseline 5.3).
- **Daily volume trajectory:** $\sim 1\text{--}23$ posts/day (Aug 20–Sep 4) $\rightarrow$ **117** (Sep 5) $\rightarrow$ **255** $\rightarrow$ **395** (Sep 13–14, the peak) $\rightarrow$ gradual decay through 157/91/58/55/80 (Sep 15–19) $\rightarrow$ a second, smaller bump to **125** $\rightarrow$ **228** (Sep 20–21) $\rightarrow$ drop-off to 18 (Sep 22).
- **Interpretation:** The pattern of a very sharp spike followed by decaying volume over 5–7 days, then a secondary smaller spike, is consistent with (a) an initial triggering event/news cycle, (b) a sustained "fan war"/commentary tail as videos keep collecting comments, and (c) a second flashpoint around Sep 20–21 (possibly a follow-up statement, event, or new video going viral) reigniting engagement before it fades again.



Hourly data and spikes

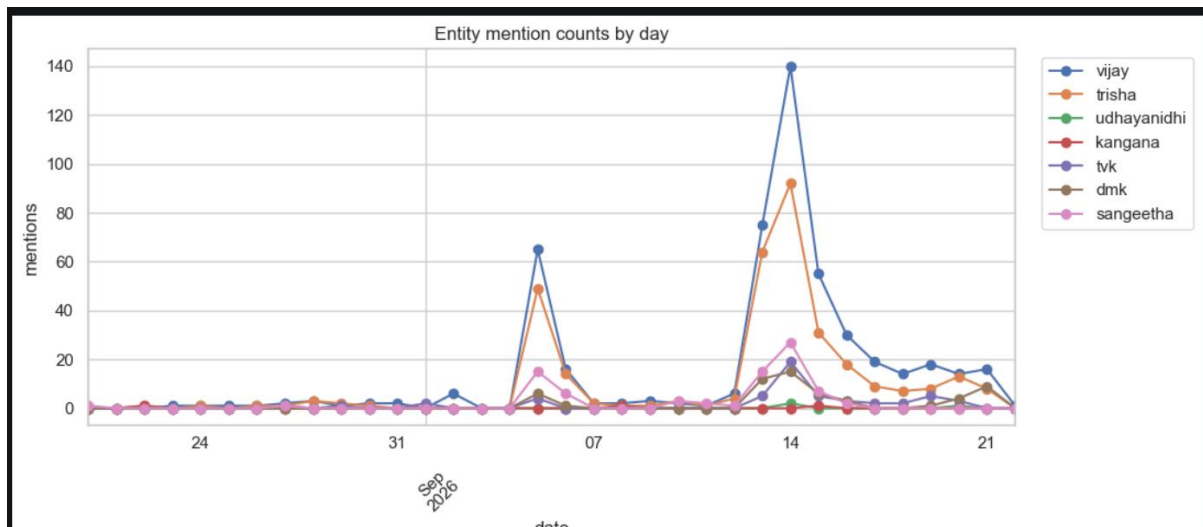


Day-wise data and spikes

5. Topic / Entity Analysis

- Round 2 topic-category model:** Labels (Community_Discussion, Technical_Issues, Feature_Feedback, Account_Security) were designed for a *product-feedback* dataset in Round 2, not celebrity/political commentary. As expected and flagged in the notebook itself, the overwhelming majority of posts fall into Community_Discussion by default this label should be reported as evidence the model's category schema doesn't map well to this domain, **not** over-interpreted as a substantive finding.
- More informative signal direct entity keyword counts** (regex word-boundary match on cleaned text), tracked daily for: Vijay, Trisha, Udhayanidhi, Kangana, TVK, DMK, Sangeetha.

Entity	Peak day	Peak count	Pattern
Vijay	Sep 14	140	Dominant entity throughout; tracks the overall volume curve almost exactly
Trisha	Sep 14	92	Second most-mentioned; consistently ~60–70% of Vijay's count
Sangeetha	Sep 14	27	Near-zero before Sep 5, then rises in lockstep with each spike (15 → 27 → 7)
TVK	Sep 14	19	Political-party angle enters conversation only from Sep 5 onward
DMK	Sep 13	15	Similarly absent early, appears only once the story politicizes
Kangana	scattered	1	Negligible occasional off-topic crossover mention
Udhayanidhi	Sep 14/20	1–2	Negligible, minor mentions



Characters involved analysis with their spikes

- **Interpretation:** Before Sep 5, entity mentions are near-zero across the board this is a "quiet baseline" period. From Sep 5 onward, **Vijay, Trisha, and Sangeetha rise together**, and **TVK/DMK enter the conversation for the first time**, indicating the controversy shifted from a purely personal/relationship story into a **political** one (tying Vijay's alleged relationship to his party, TVK, and rival party DMK) by the time of the Sep 13–14 peak. This is the clearest topical signal in the dataset and should anchor the "what people are actually discussing" section of the report over the generic Round 2 topic labels.

6. Trigger Explanations

- **Method attempted:** The notebook cross-references each flagged spike/shift date against gdelt_news rows (news headlines) collected by scraper.py --sources gdelt, checking a ± 1 day window around each date.
- **Result:** The gdelt_news source returned **zero headlines for every flagged date** the dataset in this run contains only the youtube source (8,854/8,854 records), meaning the GDELT collection step either wasn't run or returned no matches before this notebook was executed.
- **Consequence for the report:** External news corroboration is **not available from this pipeline run**. Trigger explanations must instead be inferred qualitatively from the comment text itself:
 - a) Translated sample comments around the spike dates reference a **political conference and "foreign leaders' visit"** hosted by the "Prime Minister," alongside direct comparisons of Vijay/Trisha's wealth to ordinary citizens suggesting the story was being amplified alongside, or in contrast to, unrelated political news cycles.
 - b) Comments explicitly invoke **DMK and TVK** by name at the same time entity counts for those terms first appear (Sep 5) and peak (Sep 13–14), supporting the interpretation that a specific political statement, rally, or DMK/TVK exchange (rather than a purely entertainment-press story) triggered the two largest spikes.
 - c) The reappearance of a smaller spike on Sep 20–21 without a corresponding rise in DMK/TVK mentions suggests this second wave was driven more by continued fan/celebrity-gossip engagement (e.g., a new video or clip recirculating) than a fresh political development.

- **Recommended fix for future runs:** widen --gdelt-timespan on the next scraper.py run, or add a second news/press source, so the "Reasons behind observed changes" section can be grounded in verifiable headlines rather than inferred from comment content alone. This should be explicitly stated as a data-collection limitation in the final report.